

FPGA experiment report

Task: FPGA 4

Student 1 name: _____ ID: _____

Student 2: name: _____ ID: _____

Part 1 - Status update

Fill in the following table according to your progress at the time of the submission. Each row should refer to one of the modules together with its simulation.

The status column should be filled with one of the following:

- Done - If both are written and simulation passed.
- Partially done - If you started writing the module and simulation but it's not working properly.
- Not started - If you didn't get to that part of the task.

When the status is partially done, please use the notes/comments column to explain in a few more words what is the status of the module and what is the status of the simulation.

Fill here:

Module name/Step	status	notes/comments
Control		
Debouncer		
Stopwatch		
Elaborated Design		
Synthesis		
Setting up constraints		
Implementing the design		
FPGA programming and debugging		

Issues:

In this section please describe the current issues you are facing.

- Per each module and simulation, if you receive any errors at the time of submission, please describe which errors and what they mean.

Fill here:

- Control:
- Debouncer:
- Stopwatch:
- Elaborated Design:
- Synthesis:
- Setting up constraints:
- Implementing the design:
- FPGA programming and debugging:

Part 2 - Simulations and answers

In this section, please add the images for each simulation as described in the report. Remember to use annotations and explain in your own words what is shown in each simulation.

In some of the tasks in the report, there are also questions that you should answer here.

Make sure to include images of all simulations running at the time of submission, even if they are not correct.

Fill here:

- Control:
- Debouncer:
- Stopwatch:
- Elaborated Design:
- Synthesis:
- Setting up constraints:
- Implementing the design:
- FPGA programming and debugging:

Part 3 - Questions to fill and answer

The following are all the items from the *Experiment FPGA 4 – Dual Stopwatch* assignment that explicitly ask for content in the progress report. Answer each one below (add waveforms, screenshots and explanations where requested).

Task 2 – Control

- (i) Provide an annotated waveform of your simulation. In addition, take a screenshot of the Tcl console window showing the “Test Passed” message and the “\$finish called...” message. Explain what is shown in the simulation.

Answer / images:

- (ii) The FSM is presented as a Mealy state diagram. Explain whether this FSM is a Moore FSM.

Answer / images:

Task 3 – Debouncer

- (c) Discuss the tradeoff between choosing high and low values for the counter, and how this is related to the low-pass-filter debouncing approach we saw in the previous experiments.

Answer / images:

Task 4 – Stopwatch

- (a) Design a block diagram for the Stopwatch module. Use the previous designs as instantiated modules (no need to show the internal logic); show and explain any other logic and connectivity.

Answer / images:

Task 5 – Elaborated Design

- (b) Provide the screenshot of the elaborated design schematic.

Answer / images:

Task 6 – Synthesis

- (i) Provide a screenshot of the synthesized design schematic.

Answer / images:

- (ii) Describe the differences that appeared in the synthesized design compared to the elaborated design.

Answer / images:

- (iii) Describe the issues you encountered during synthesis and how you solved them (errors, critical warnings, and warnings).

Answer / images:

Task 8 – Implementing the design

- (i) Provide a screenshot of the design timing summary where you must exhibit positive slack both for setup and for hold times.

Answer / images:

- (ii) Specify the lengths (in nanoseconds) of your critical setup path (i.e. the *path length* with the lowest setup slack – do not just specify the slack value).

Answer / images:

Task 9 – FPGA programming and debugging

- (c) After your design was successfully approved, remove the debug core, re-implement the design, and provide a screenshot of your Project Manager → Project Summary screen including post-implementation Timing, Power, and Utilization details (table view).

Answer / images: